



POLITECNICO
MILANO 1863

Digital and Interaction Design

Final Design Studio

A.Y. 2024/25

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REPORT
January 2025

ECCO

Enhance family
bonding in the
Museum

In 2035  mnm



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Background and Objectives

Seeking meaning of the museum

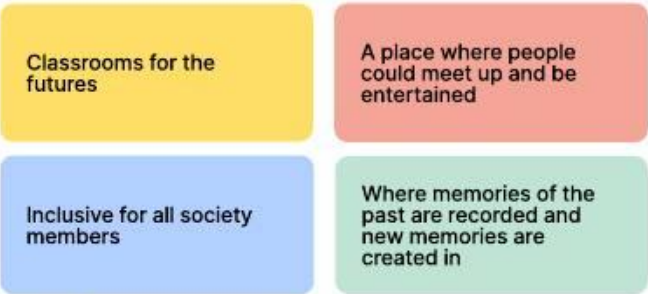
The general objective of this project is to explore the meaning of museums and how their meaning would evolve in the future to better align with people mindset and lifestyle.

The word “museum” originally comes from the Greek, meaning “the holy place of the Muses, who are the goddesses of arts and science.”

Over the centuries, the concept of museums has evolved. According to Wikipedia, a museum is defined as “an institution dedicated to displaying and/or preserving culturally or scientifically significant objects.”

Meaning of museum in the future

The vision we developed in the initial phase of the project for the future of museums is:



[Fig01] Meaning of the Museum in the Future, ideated by Authors.

Research & Observation

Design Brief

Designing **Age-Appropriate, Engaging, and Interactive** Experiences for Children and Parents in the Natural History Museum of Milan in 2035 through **Digital and Phygital** Technologies.

Desk Research

In the first phase of the project, to better understand family interactions in museums, their needs, challenges, and vision for the future of museums, we conducted desk research, stakeholder analysis and analyzed case studies from museums around the world.

This approach helped us explore recent developments and interactive practices. The most valuable insights on the interaction are summarized in the following:

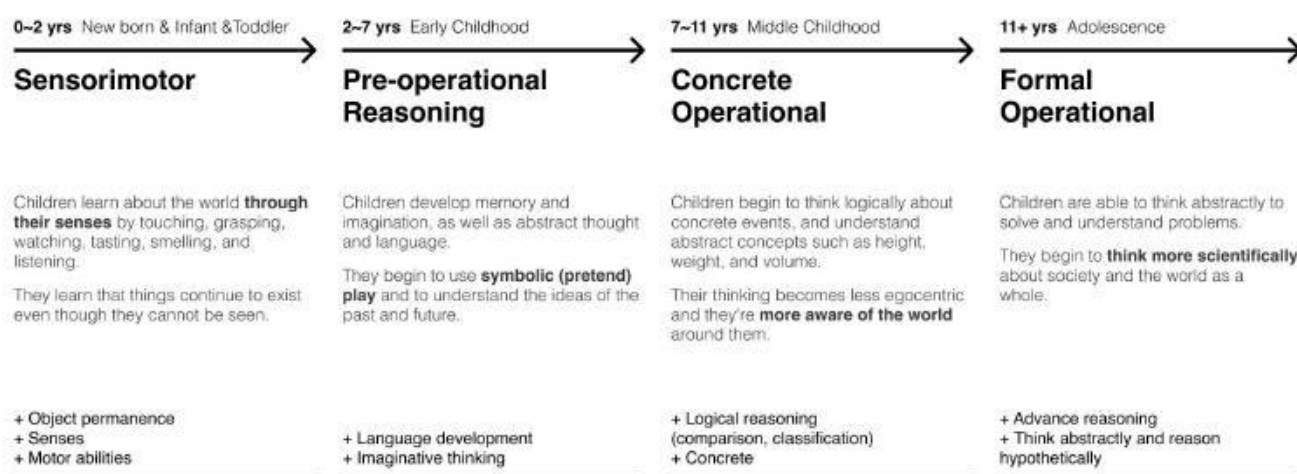


[Fig02] Visual Representation of Key Interaction Types Based on Authors' Insights.

The studies also reveal that families tend to spend more time with each other than focusing on the exhibition itself and the objects on display, suggesting that museums can provide a special environment for family communication. For this reason, an important goal for an effective exhibition is to encourage dialogue between family members.

Children Cognitive Process based on their age

Jean Piaget's Theory suggests that children move through four different stages of learning. His theory focuses on understanding how children acquire knowledge, and on understanding the nature of intelligence.



[Fig03] Summary of the Cognitive Process by Authors.

Parents Interactions with kids in the museum

Studies categorize parents' interactions with their children into four main types:



Research findings suggest that the most effective way to address parents' needs during museum visits is to **create shared experiences** for families. By providing **structured guidance** and museum's **mediated learning**, museums can enable parents to engage with exhibits alongside their children, reducing the risk of misinterpreting information and enhancing the overall experience for both.

Children favorite interactions in the museums:



Role-Playing



Physical Activities



Interactive Storytelling

To gain a deeper understanding of interactions within the Natural History Museum, we conducted a series of structured observations. These focused on the behaviors, preferences, and engagement patterns of parents and children.

Observation Results

Observations revealed the following problems in the natural history museum of Milan:



Monotonous Exhibits



"Not child-friendly"



"Parents as museum interpreters"



"Hard to understand the content"

Content

The museum's content is often **too complex** for children to **understand independently**.

Exhibits fail to incorporate storytelling techniques, reducing the effectiveness of **information delivery**.

Design

Some halls lack engaging interactions, leading to **monotonous** visitor experiences.

Certain halls are not designed with children in mind, featuring exhibit heights that are too tall for them to view or read.

User Interaction

Parents frequently take on the role of interpreters, explaining museum content to their children.

Existing interactive experiences are underutilized due to the absence of explanatory banners or clear instructions, making them **challenging to navigate**.

Conclusion

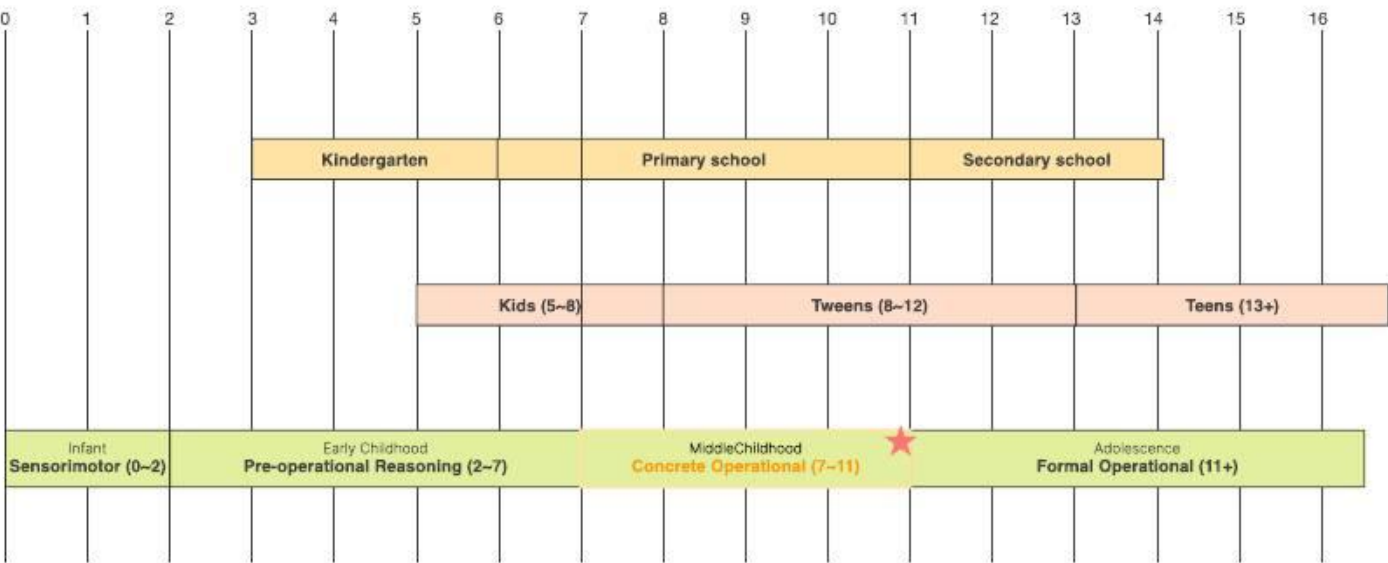
Families primarily visit museums to spend quality time together and share meaningful experiences. Research shows that collaborative learning methods, such as physical games, role-playing, and Q&A sessions, are the most effective ways to engage families. However, children's distinct cognitive learning styles require **age-appropriate** interaction design.

At the Natural History Museum of Milan, families face challenges in engaging effectively. Key issues include complex content, monotonous exhibits, and a lack of interactive elements for children. Additionally, parents often act as interpreters because the exhibits are not designed with younger audiences in mind.

Understanding the User

User Target Group

We chose children aged 7–11 as our user group because they are developing logical thinking, understanding abstract concepts, and becoming more aware of the world. This makes them highly receptive to engaging, hands-on learning experiences, aligning with our goal to create interactive exhibits that foster curiosity and meaningful family interactions.



[Fig04] Juxtaposition of the Cognitive Development Process and the Italian Education System by Age Group: A Chart Designed by the Authors.

Semi-structured Interview

In our study, we interviewed **30 families**, the primary visitor group identified for the museum consists of children aged **7 to 11** and their parents, who are typically between **35 and 55 years old**. Before we carried out the interview, we developed case studies in order to generate useful interview questions and obtain our insights. (specific questionnaire and case studies, please see appendix)

Case Studies

Exhibition	Exhibition	Exhibition	Exhibition
+ A microscopic view of the macroscopic world	+ Experiencer rather than witness: to be a part of this journey	+ “Non-everyday”: uncover hidden	+ Cross-platform: before and after museum
+ Impersonal perspective		+ Immersive environment	
“Follow Bunji, an Australian green sea turtle, on her first solo journey across the open ocean as she meets incredible animals along the way.”	“Audiences will encounter species that are adapted for the cold—some long extinct, others still alive today—and learn how a changing climate could alter the face of our planet in the future.”	“Explore nature’s hidden realms —including some that are too fast, too small, or too slow for the human eye.”	“Visitors can share the snowflake designed by themselves on social media as a digital keepsake of the exhibition and access it online.”



Program	Program	Event	Program
+ Classroom + Field Trips	+ Experimental Field + Tiniest curators: let them modify what they what	+ Scientific bar + Inclusive: for adults, be a kid again	+ Hospital: museum prescription
"During a field trip visit, students creatively participate in the practices of scientists and engineers to help students engage in science opportunities that they might not have in the classroom."	"Rotating monthly themes, STEAM activities, and modifications to exhibits to make them more accessible to little ones. Cooperated with UCSB's Child Studies Lab. Facilitate an activity they have designed for toddlers based on scientific concepts."	"Why should kids have all the fun? Now it's time for the grown-ups to play and be curious. We have events on tap for you to enjoy without any kids in sight. Beer, wine and non-alcoholic drinks available for purchase. All guests must be 21 or older."	"Psychiatrists in one of the city's largest hospitals have been able to offer patients "museum prescriptions", a free visit with a few friends or family members to discover one or more of Brussels' cultural institutions."
			

Insights

Based on case studies and literature review, we generated our insights for discovering what's the future of the Natural History of museum in 2035?

Why they go to the museum

Parents' reason :

1. Expose children to knowledge beyond daily learning.
2. Increase children's interest in school subjects through the museum.
3. Enhance parent-child interaction and spend time together.

Children's motivation:

Enjoy the museum exhibition content and want to **learn** more.

What are the focal issues now?

+ Observation

1. Traditional out-of-date showcase
2. Users' behavior (Children & Parents)
3. For children: not child-friendly
4. For parents: "information interpreter"
5. Cognitive overload
6. Not intuitive
7. Lack of affordance & signifier
8. Lack of communication within ecosystem

Favourite Interaction activities



Physical Game



Live show with singing and toy



Storytelling



Interactive digital installation



VR experience



Drawing

What's experience in the museum

Most Popular Halls and Exhibits



1. Mineralogy



2. Paleontology



3. Natural History of Human

Interesting Reason:

1. Visual appeal
2. Content Interest
3. Unusual perspective and interactive experience

Less Engaged Halls



1. Vertebrate zoology



2. Entomology

Boring Reason:

1. Content hard to understand
2. Poor visual effects

User Interaction at the museum



1. Physical Game



2. Live show



Text description



Storytelling

Expectations:
Interactivity
Creativity
Participation

Children's favorable interaction way in the museum.



Interactions between Parents and Childrens in the museum now.

Guided Learning: Parents lead children to explore exhibits and encourage observation. (emotional sharing)

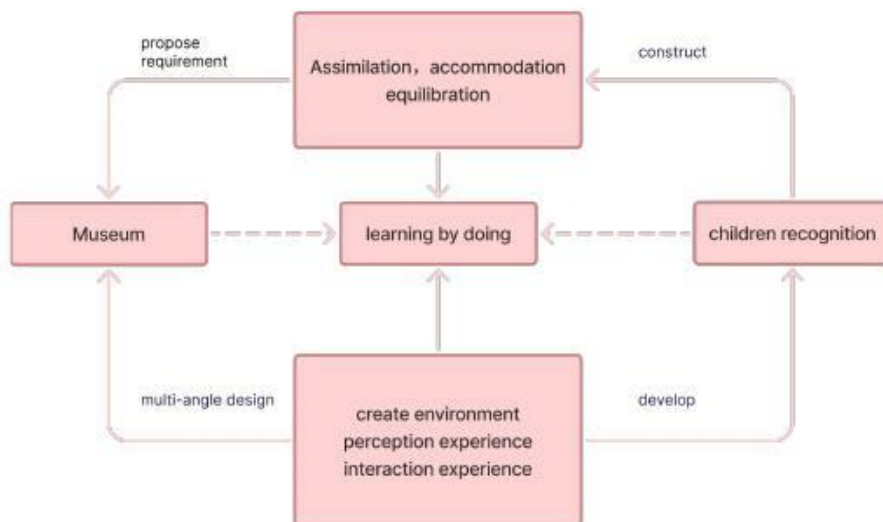
Explanations: Parents explain the exhibits, providing additional context.

Q&A: Children ask questions, and parents help clarify or provide answers.

Encouraging Independence: Parents motivate children to learn through their own exploration.

Interactive Challenges: Parents pose questions or use treasure-hunt style games to engage children.

Learning way in the museum



Families, particularly children and parents, preferred a **collaborative learning experience** in the museum. They enjoy working together to explore and engage with exhibits, rather than relying on a museum guide to lead them through the experience. This approach fosters interaction, strengthens family bonds, and makes learning more enjoyable and memorable for both parents and children.

[Fig05] Learning way of children in the museum

Problem Statement & Context

Problem Statement

When families visit the museum, parents act as **mediators** between the exhibits and their children, but they often **struggle to simplify complex information** and answer their children's questions, making the experience **less enjoyable and disrupting** family engagement.

User Persona

In order to better understand and address the unique goals, preferences, and pain points of different family members, ensuring a more inclusive and tailored museum experience for all.

We create three user persona based on the needs and expectations of families in the museum, outlined in the previous chapter:

- Child—Explorer Ellie
- Mother—Caring Clara
- Father—Busy Bruno

These personas were informed by Falk’s Museum Visitor Motivation Identities and reflect the diverse motivations and challenges families face during their museum experiences.

Signal of Change

Rise of Edutainment

The growing trend of combining entertainment with education is pushing museums to create more engaging, interactive experiences for children.

Digital Transformation in Museums

Museums are increasingly using digital tools such as AR,VR and interactive displays to modernize their exhibitions and appeal to families interested in technology

Family-Centric Learning

There is a shift towards family learning experiences where parents and children can explore and learn together,which is h line with the increase in collaborative learning activities

User Personas

Exploring Ellie

Child



Ellie Mossi
7 years old

Primary school students (Second Grade)
Milano

"I want to have fun while learning! I love it when I can touch things and see how they work—like going on an adventure in the museum!"

Description

She is a curious child who thrives on discovery and exploration. She enjoys learning about animals, science, and history. She enjoys spending time with her family on weekends.

Traits

Education Background: ☐
Knowledge level of school science: ☐
Tech-savvy: ☐

Needs

- Variety of Stimuli: Engaging with museum content through multiple senses.

Expectations

- Get knowledge about animals and fossils.
- Means to discover new things and play while learning.

Engagement Style

- **Discoverer**—she wants to explore thoroughly, ask questions, and learn about the exhibits. She enjoys spending time at each display, get information and engaging with anything that catches her interest.

#Explorer #Curious #Discovery #Adventure #Learning

User Personas

Caring Clara

Mother



Clara Mossi
38 years old

Chef in a small cafe
Milano
Married to Bruno, mother to Ellie (7)

"I believe the best learning happens through doing, where my daughter can explore and grow her curiosity."

Description

She is a nurturing mother who believes in the power of experiential learning. She may not have a high level of formal education, but she prioritizes practical learning experiences for her child.

Traits

Education Background: ☐
Knowledge level of natural sciences: ☐
Tech-savvy: ☐

Needs

- **Supportive Environment**: Looks for activities that encourage family bonding and allow for active participation.
- **Hands-On Learning Experiences**: Seeks activities that engage children through tactile and practical involvement.

Expectations

- **Practical & Educational Activities**: Values learning experiences with real-world applications that inspire curiosity, creativity, and critical intelligence in Ellie.
- **Simple & Intuitive Navigation**: Prefers activities that are easy to navigate, as she is less tech-savvy.

Engagement Style

- **Swimmer**—she engages at a moderate level, moving through exhibits while ensuring her children are involved and learning. She balances her own interest with the experiences of her kids, looking for ways to facilitate their enjoyment.

#Facilitator #HandsOnLearning #InvolvedParent #FamilyTime

User Personas

Busy Bruno

Father



Bruno Mossi
40 years old

Project Manager
Milano
Married to Clara Mossi, father to Ellie (7)

"I work long hours, but Sundays are family days. I want experiences that create lasting memories and help my kid learn."

Description

He is a hardworking project manager who often struggle to find work-family-life balance. However, he knows the importance of spending quality time with his children.

Traits

Education Background: ☐
Knowledge level of school sciences: ☐
Tech-savvy: ☐

Needs

- **Structured Yet Flexible Experiences**: Prefers activities that provide clear structure while allowing flexibility to accommodate his limited time.
- **Quality Family Time**: Values educational and engaging activities that foster his child's development during his limited free time.

Expectations

- **Didactic & Engaging Exhibits**: Enjoys experiences that blend education with fun, especially in natural sciences, to capture Clara's interest.
- **Tech-Savvy Tools**: Desires an engaging, tech-driven environment where Ellie can explore and learn while feeling connected.

Engagement Style

- **Skimmer**—he wants to see key highlights without diving deeply into content. He just want to make his daughter happy.

#ExperienceSeeker #FamilyTime #Efficiency #WorkshopFather

[Fig06] Persona Cards, designed by Authors

UVP



Our idea enables **families** visiting the museum to have a **shared experience**, explore and learn together, in order to **improve family bonding** and create lasting memories.

HMW

How might we create an engaging and collaborative museum experience that **fosters family bonding**?



Ideation and Design

Concept Ideation

After identifying our target users, we explored new realities and meanings for the natural history museum experience. The goal was to enhance engagement for parents and children while fostering a meaningful connection to the exhibits. This process led us to **identify patterns of change informed by meaning**. Through brainstorming sessions, we generated five possible ideas to address these opportunities:



Spinny Stories: A **collaborative game** where family members use **interactive sphere devices** to engage with museum exhibits through tactile feedback, **unlocking layered information** and **triggering cooperative tasks**.

1 2



Dino Evolutionary Network: By turning a handle, users can **highlight dinosaurs** in evolutionary order, with a screen **providing a simple story** for children and advanced details for parents.

3 4



EcoQuest: A Journey Through Time: An interactive adventure combining time travel and ecological exploration, where families use a **time-travel passport** to learn about the evolution of nature.



Unboxing the museum: A mobile app that uses **interactive quizzes** and timelines to explore the history of dinosaurs, encouraging **learning through fun, rewards, and digital engagement**.

5

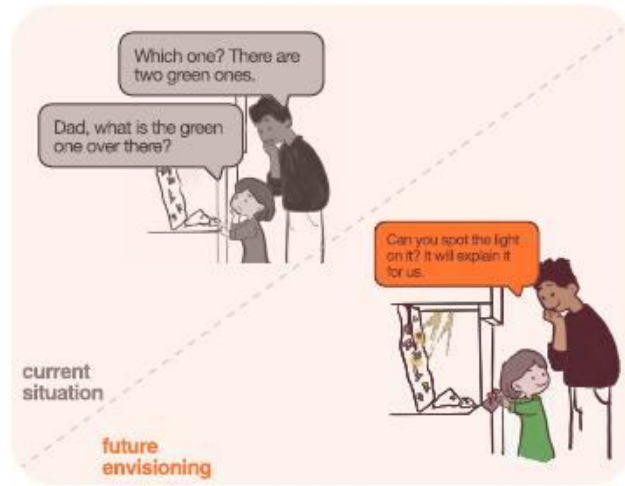
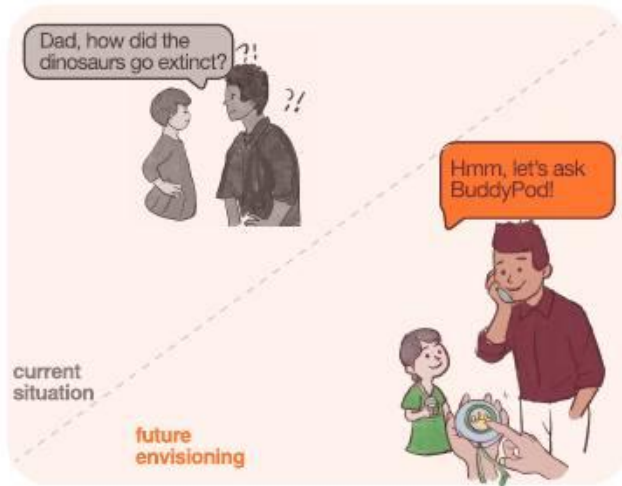


Tomorrow's Laboratory: A multi-faceted approach to empower children as global citizens, blending empathy, critical thinking, and interactive exhibits like **role-playing** as scientists and fighting sea pollution through **personalized AI assistance**.

After evaluating and refining these ideas, we realized **family bonding** as the **core meaning** of our project, emphasizing the emotional connections families can form while learning together. To create a seamless and memorable experience, we structured the visit around three key steps: **pre-visit, during visit, and after visit**. These phases were designed to guide families through the museum experience, with a boundary object serving as a shared tool that connects the different elements and supports the family's learning journey.

Two Primary Scenarios

We merged and synthesized the five initial ideas to create two refined scenarios, each incorporating the most valuable features from all the concepts.



Scenario 1-Interactive Storytelling

Focuses on an **AI-powered companion** that guides families through the museum, enhancing learning through **voice interactions, storytelling, and quizzes**. It promotes collaboration between parents and children in a fun and educational way.

Scenario 2- Interactive Diorama

Features a **multi-sensory diorama** where families can engage with exhibits through sight, sound, and touch. **Collaborative games** encourage family bonding while exploring the museum, making the experience more interactive.

S1 - Storyboard



S2 - Storyboard



Iterative Design and Testing

MVP Test

To validate our ideas, we conducted MVP tests to determine whether the defined problem truly exists and to observe how users experience it. The goal was to understand the interactions between parents and children during the museum visit and explore whether two scenarios effectively addresses their needs.

Before the MVP tests, user journey maps were prepared for each scenario to outline the expected flow of interactions. During the test, participants acted as family members, taking on the roles of parents and children to simulate real-life museum experiences.

(specific UJM see appendix)



Both tests focused on key questions:

1. Does Scenario 1/2 enhance co-learning and family bonding?
2. How do parents and children collaborate or engage during the learning process?
3. Do participants face challenges in understanding the exhibits or answering children's questions?

After conducting the MVP test, we selected **Scenario 1** for further development as it proved to be the more promising option. Because:

Education + Entertainment

- + Provide collaborative learning
- + Promising education value while having fun.

Better User Control

- + The object's shape is intuitive.
- + Easy for children to use and control.

Promising Family Bonding

- + Direct interaction between parents and children.
- + Appeal to both parents and children.

Feasibility: S1 > S2

- + S1: Technology feasible with current AI industry; Easier to maintain as it is not integrated into collection showcases.
- + S2: It requires renovation of existing diorama showcases and it may cause issues in terms of specimen conservation and maintenance.

Scalability: S1 > S2

- + S1: The object is independent and therefore potentially versatile and can be used in multiple scenarios.
- + S2: The installation is limited to diorama this single form, so it can't be adaptive to other kinds of collection.

Iteration of Scenario 1

Evolving the Experience

While Scenario 1 showed promise, further testing revealed that the question-and-answer activity could become frustrating over time. This led us to reassess interactive storytelling as the core activity.

Discovering Natural Interactions

Our refinement process highlighted that the most meaningful family interactions stem from shared activities—collaborating, co-creating, or simply enjoying moments together. Recognizing the importance of shared responsibility and mutual support in family dynamics, we integrated these elements into the museum experience. By reflecting familiar activities like problem-solving, playing games, or storytelling, we aimed to foster deeper connections and engagement.

Scenario 1

quiz-based
interactive storytelling



ECCO

voice-interactive family companion,
transforms museum visits into
engaging, co-creative storytelling
adventures

Potential Competitors

Museum usage

They have variety of contents but it doesn't provide co-learning.

Actionbound

An interactive app-based treasure hunt game and guided walks.



Google Arts & Culture

An online platform using tools and technologies for cultural organizations to showcase their cultural treasures.



At-home usage

While they offer high-tech features, they are designed for at-home use and target only children, limiting in museum settings.

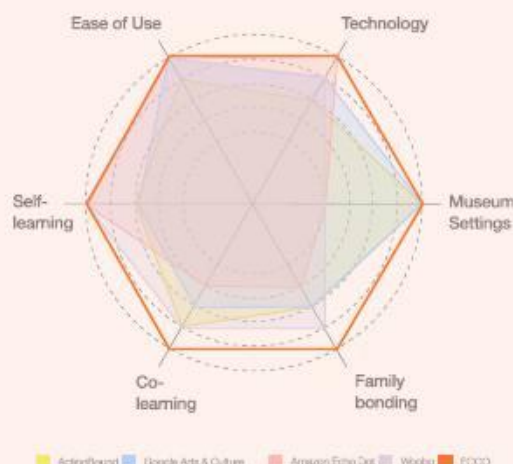
Amazon Echo Dot - Kids

A smart speaker with Alexa, made for kids (not a toy), delivering clearer vocals and rich, vibrant sound.



Woobo

An interactive, educational robot companion with a kid-friendly question and answer software and update-able content.



Concept Overview

Ecco

Ecco turns museum visits into **fun, interactive** adventures for families. **NFC-enabled** cards, **linked to specific exhibits**, let visitors create **personalized stories**, solve mysteries, and **earn badges** for their **museum passports**. These stories grow with each visit, offering new adventures every time. An AI-powered companion enhances the experience, turning passive visits into engaging, collaborative journeys. It frees parents from a "teacher" role and captivates children with immersive storytelling. By combining NFC technology and AI, Ecco **strengthens family bonds**, encourages **personalized learning**, and helps museums **retain visitors**.

The Metaphor : Scouting

To capture the spirit of adventure and exploration, we drew inspiration from scouting in our experience. Just like scouts on a journey of discovery, families embark on a shared learning adventure through the museum. The **Family Passport acts as a guide**, providing all the necessary instructions and guidance to help families navigate through various exhibits. **Ecco is designed like a Swiss knife**—versatile and multifunctional—equipping families with the tools they need to uncover mysteries along the way. With the freedom to choose their own path and solve challenges, families become active participants in their journey. Just as scouts collect badges for their achievements, families earn physical badges as symbols of their progress. These badges not only encourage a sense of accomplishment but also strengthen their connection to the museum, creating a deeper and more rewarding experience.

System Design

Closed Card Sorting

To refine the interaction flow and organize touchpoints, we conducted closed card sorting. This method allowed us to align the system's architecture with user expectations by exploring how participants categorized interactions such as profile setup, storytelling, and badge collection.

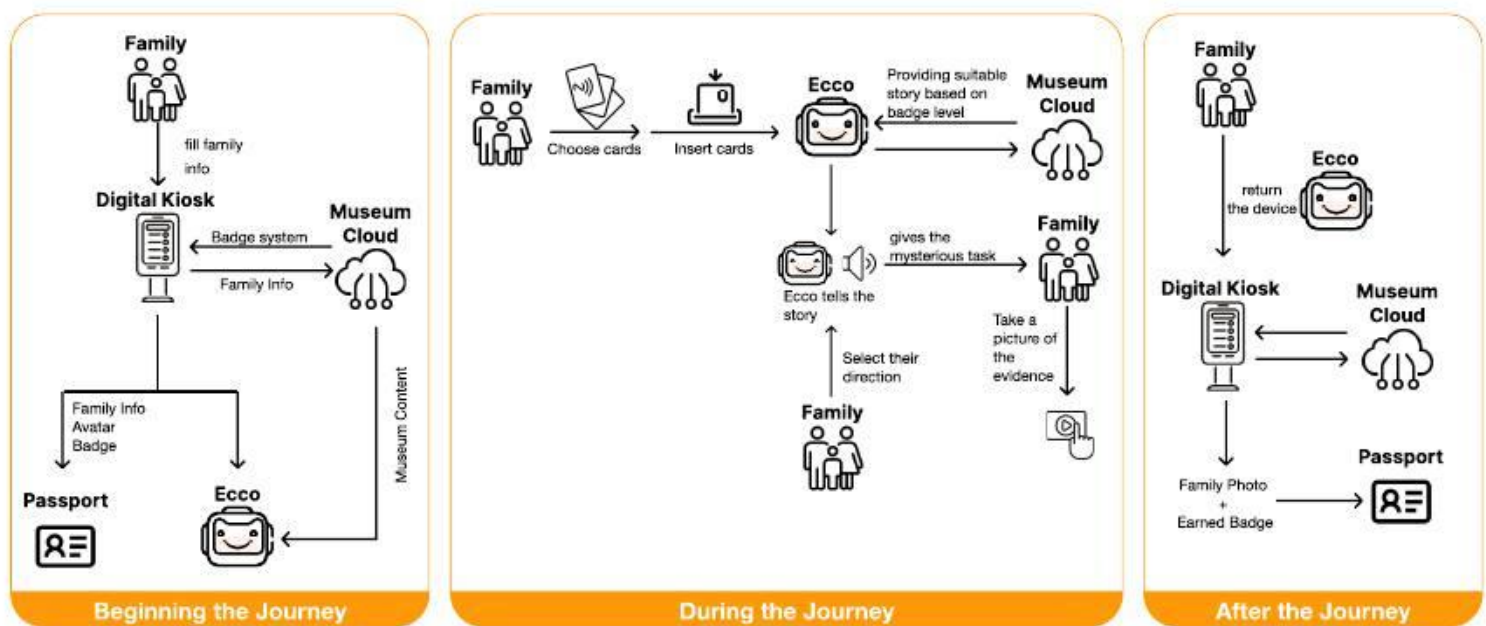


Key Insights and Findings

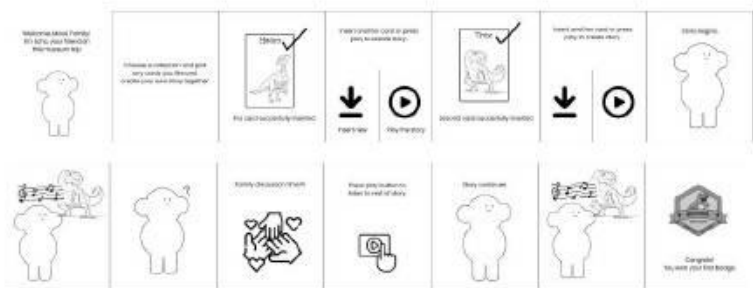
The majority of participants grouped the elements as expected, as shown in the chart above. However, there was a minor issue regarding the understanding of the passport's function. Some participants grouped the museum map under the digital kiosk instead of the family passport. To address this confusion, we propose providing a short, instructive video that explains the ecosystem before users purchase museum tickets. This video would not only clarify the system but also serve as a promotional tool and the first point of engagement with users.

System Architecture & User Flow

The card sorting test helped prepare the user flow and system design to align with user expectations. Every interaction and step was thoughtfully planned to guide users smoothly through the pre-visit, during-visit, and post-visit stages as as outlined below.



Lofi

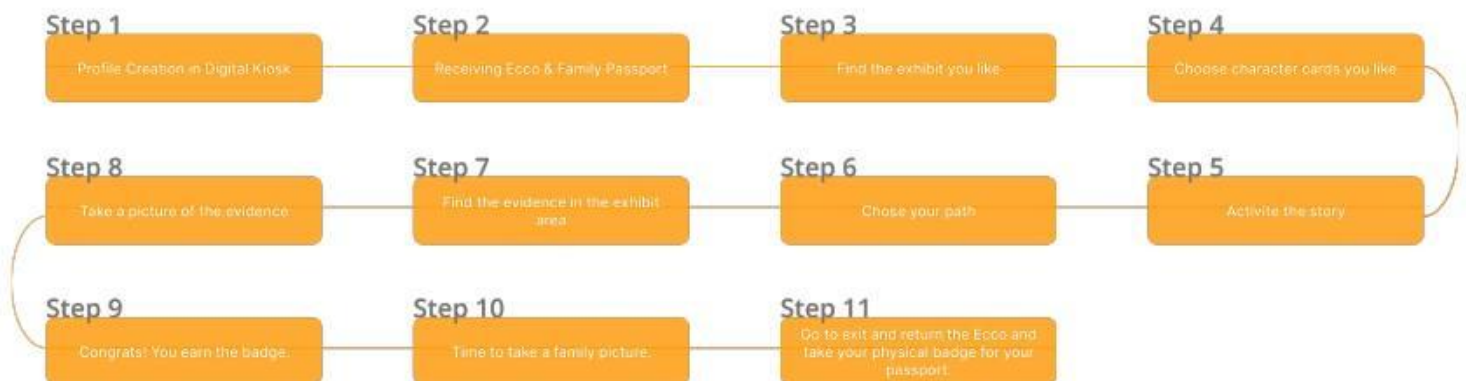


Testing and validation

User Test Plan

Participants	Around 52 users (for quick iteration and qualitative data)
Location	EDME LAB (Polimi)
Time	13:00, 04.12.2024
Goal	1. Testing Family Bonding: Assess how the experience fosters meaningful connections through shared activities. 2. Evaluating Family-Friendly Design: Determine the inclusivity and accessibility of the approach for diverse family groups. 3. Engagement with Interactive Storytelling: Measure how immersive narratives captivate and involve participants. 4. Creating Memorable Experiences: Test the ability to deliver lasting and impactful moments for families. 5. Identifying Strengths and Weaknesses: Analyze the experience to highlight its successes and areas for improvement.
Roles	Participants role-played as parents and children to test the museum's interactive experience. The evaluation utilized three methods: open-ended questions, a feedback wall, and the Subjective Experience Questionnaire (SEQ).

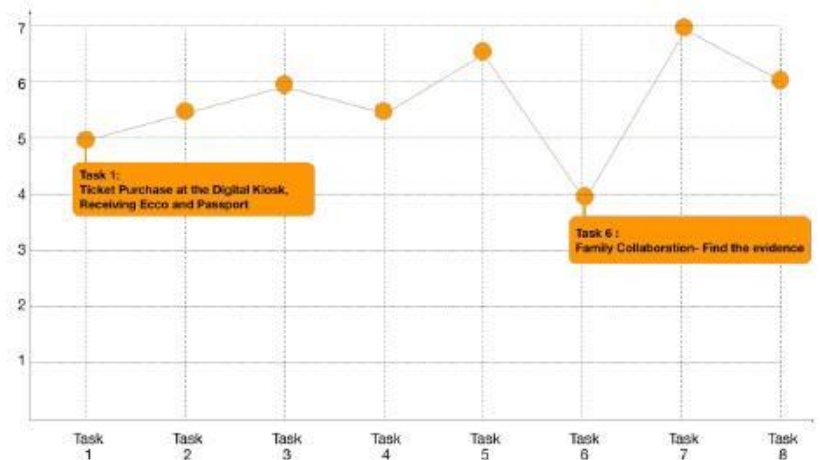
Task Flow	To ensure the effectiveness of the entire experience, we divided the main interactions into task flows and instructed users to complete each task based on the provided guidelines. The key tasks identified are listed below.
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Test Results

SEQ Result

The SEQ results indicate that most users faced challenges in finding the evidence related to the story (Task 6). Additionally, users had difficulty retrieving the product and passport from the digital kiosk. Among the tasks, the most popular one was taking a family photo with Ecco, which directly aligns with the product's focal goal of fostering family bonding. Furthermore, providing users with the opportunity to play a role in the story and choose its path was highly engaging (Task 5).



Feedback Wall Result



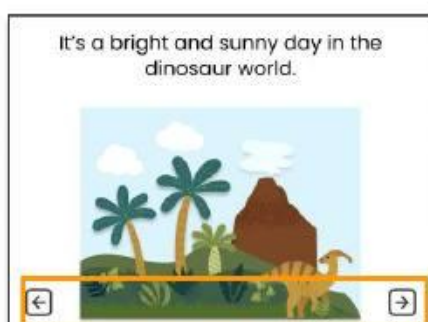
Analysis of the feedback wall responses reveals that, in answer to the question about what they liked most about the experience, the majority of users mentioned feeling a **stronger connection** with their child. Additionally, users noted that the experience effectively **fosters family engagement**.

Key findings

In the following the most mentioned challenges during the test is mentioned :

Misleading Buttons

The next buttons are too intuitive to press, causing confusion for the user.



User Control and Freedom

Users want the option to stop the current story and choose another collection. The system currently doesn't offer control over story progression, leading to frustration for users.

System Visibility

Users need to track their progress while watching the animations, which distracts their focus.



Limited interaction with diorama



1. The evidence for solving the mystery during the story was not clearly visible to the users.
2. Users struggled to find card insert area.
3. Users were not satisfied with holding onto the passports during the test.

Prototyping and development

Refined Concept

Based on the test results, we refined the parts of the experience that required adjustments. The main revisions and possible solutions are outlined below:

Enrich interaction with Ecco

Feedback showed that users want more interactive elements, not just with Ecco but also with the diorama, to enrich their experience. To address this, we developed additional stories with varied content and interaction styles. New features include using a compass to solve mysteries and integrating scent-based interactions for a more immersive experience.

Story 1



Tyrannosaurus rex Hadrosaurus

Hadrosaurus eats a smelly plant, and Triceratops spots it, starting the journey. User helps Triceratops follow the scent and clues to find where Hadrosaurus went.

Story 2



Triceratops Hadrosaurus

Hadrosaurus eats a smelly plant, and Triceratops spots it, starting the journey. User helps Triceratops follow the scent and clues to find where Hadrosaurus went.

Story 3



Tyrannosaurus rex Triceratops

Users help T. rex hunt Triceratops' hidden eggs or protect them from the predator. Use a magic infrared camera to find the eggs before time runs out!

Story 4



Tyrannosaurus rex Hadrosaurus Triceratops

Hadrosaurus and Triceratops are hiding from the hungry T. rex. User helps them find a safe place using a magic compass to avoid detection

Design principles and guidelines

Designing for children and parents in museums requires thoughtful consideration to ensure the environment is both engaging and accessible for all ages. We adopted several principles to create interactive and family-friendly spaces that encourage exploration and learning. After conducting preliminary research on designing user experiences for families, we identified key principles to guide our design:

1. **Clear and Readable Typography:** Use large, high-contrast fonts to ensure readability for diverse age groups and individuals with visual impairments. [.]
2. **Color Palette with High Contrast:** Employ visually stimulating colors for children while maintaining simplicity for parents. [x x]
3. **Illustrations and Animations:** Incorporate playful visuals to engage children and maintain their interest. Use animated graphics or simple illustrations for exhibit instructions. Include characters or mascots that guide children through exhibits. [x x]
4. **Multisensory Visual Displays:** Ensure visual design elements are consistent for familiarity and comfort. Use consistent fonts, color schemes, and icon styles across exhibits. Create uniform visual guides for maps, brochures, and signage.
5. **Consistency Across the Museum:** Offering accessible help options, including voice assistance and camera features, ensure users can get assistance when needed.



[Fig07] Inspiring New Interactions: How Museum Apps are Changing the Game for Cultural Institutions



[Fig08] MOBILE APP : Empower visitors to shape their own cultural experience



[Fig09] The Children's Museum of Indianapolis--Card game



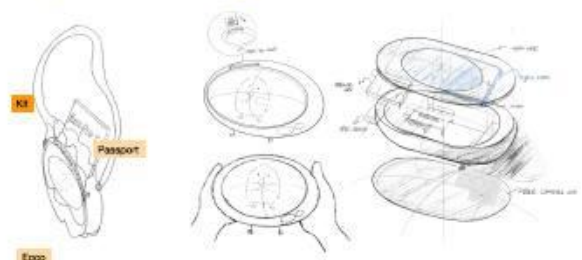
[Fig10] Actionbound--design your own game in the browser-based Bound Creator and let your visitors play it with the app

Hi-Fi Prototyping

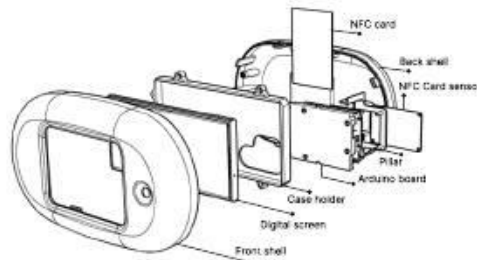
Phygital Prototype

After define the principles, we carried out phygital prototyping blends physical and digital elements to create interactive experiences. It involves **crafting final story**, developing **visuals and narration**, designing **user interactions**, and building a **product physical structure with integrated technology**. Through testing and refinement, the final prototype ensures **seamless and engaging storytelling**.

Physical Prototype



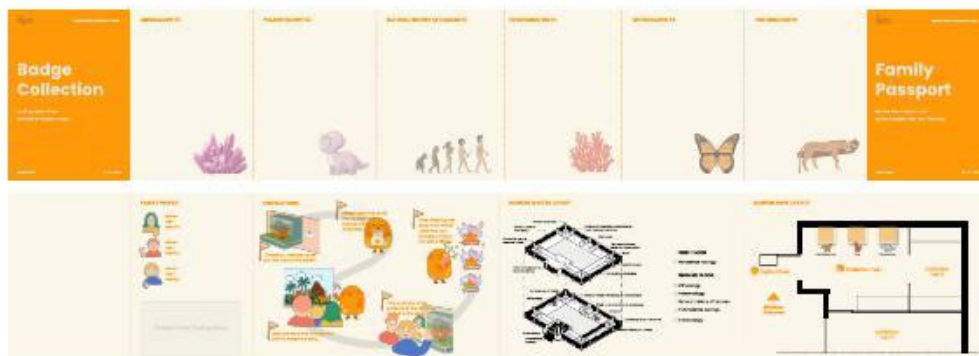
[Fig11] Ecco draft version



[Fig12] Ecco Modeling version



[Fig13] Ecco Final Version



[Fig14] Passport design

The passport serves as a keepsake, **recording badges and memories of the journey**, making every visit unique and rewarding.

Ecco Device uses **NFC technology**, an **Arduino board**, and a **digital screen**. Paired with a Museum Passport, it transforms museum visits into personalized adventures. Visitors select character cards near exhibits, insert them into Ecco to activate stories, and use their passport to **track progress** and achievements.

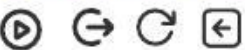
Design System

Every element, including graphics, icons, and illustrations, was carefully chosen and designed to suit our users' needs.

Typography

We chose the font "**Poppins**" to ensure readability, influenced by its large font sizes. We considered a minimum font size of **15 pt** for clarity.

Colors and Icons



Brand





Primary



Secondary



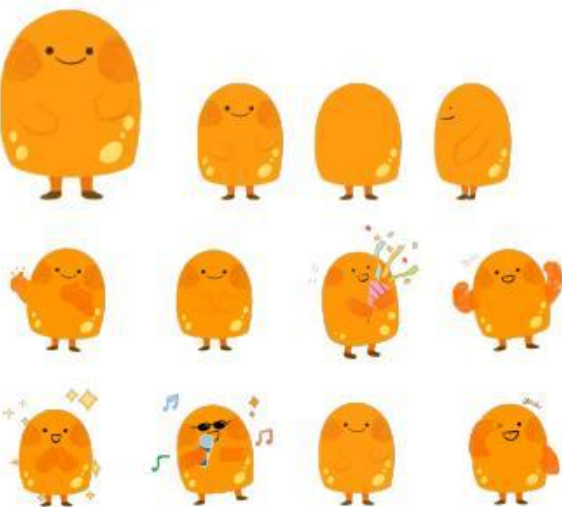
Poppins

Regular | Medium | Bold

Aa Bb Cc Dd Ee Ff Gg Hh Ii Jj Kk Ll Mm Nn
Oo Pp Qq Rr Ss Tt Uu Vv Ww Xx Yy Zz
0 1 2 3 4 5 6 7 8 9

ABCabc

IP Design



Branding

We designed the logo and tagline to embody the essence of Ecco, our family adventure service that reimagines museum experiences for the future. As an innovative platform, Ecco transforms museum experience into dynamic, personalized, creating meaningful relationships between parents and children while shaping a new way to explore and learn— stepping into the museum of 2035.



Digital Prototype



(Detailed Prototype see appendix)

Conclusion

Conclusion

Ecco aims to reimagine the **Natural History Museum of Milan** as a space that fosters **family bonding** through interactive and participatory learning experiences. By prioritizing collaborative activities and multi-sensory interactions, it strengthens the connection between parents and children within the museum setting. The integration of tools like Ecco and the museum passport enhances engagement, eases the interpretive burden on parents, and provides children with memorable, immersive experiences. This approach bridges the gap between traditional exhibits and the evolving expectations of modern families, creating a vibrant and inclusive museum experience.

Throughout the process, we experimented with diverse ideas, overturning and reshaping concepts repeatedly to arrive at the final solution (our team nickname: "Ideation Machine"). For example, we explored interactive storytelling through the Ecco device, gamified elements like badge collections, and collaborative tasks that families complete together, we search and think a lot about Scientific, Rational, Fun, Interactive. These ideas were refined through iterative testing and design, resulting in a comprehensive experience flow that accounts for every interaction point during a museum visit.

Creating a perfect dish requires numerous trials, the right ingredients, and precise timing. Similarly, this project involved extensive research, discussions, design iterations, and refinements to deliver the final outcome. Despite limitations in technology and time, we believe that the museums of 2035 will reflect elements of our vision and design.

Future Scope and Limitations

Ecco has the potential to expand by integrating advanced technologies like AR/VR for immersive storytelling and personalized AI-driven learning paths. It could also be adapted to cater to broader audiences, including younger children, teenagers, and visitors with disabilities, while incorporating contemporary themes such as sustainability and climate change. However, its reliance on digital tools presents maintenance challenges and accessibility barriers for less tech-savvy visitors. Additionally, the scalability of the system across other museums and the significant resources required for implementation may limit its immediate applicability. Despite these constraints, the project lays a strong foundation for the evolution of family-centric museum experiences and provides a new vision that creates a new value for visitors and the world.

Reference

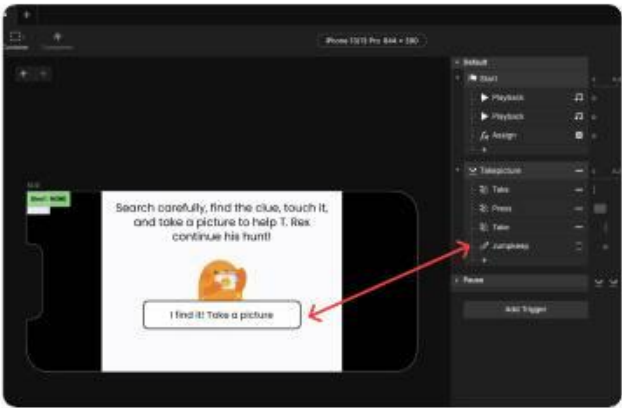
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Appendix

1. ProtoPie

Utilizing ProtoPie's "Trigger" function to control audio playback, interactions, and scene transitions.

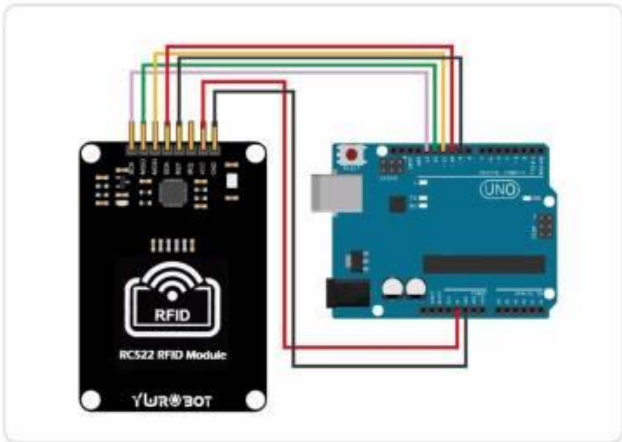
For example, when the user presses the button labeled "I find it! Take a picture," the prototype transitions to the "Take Picture" scene. In this scene, the camera component provided by ProtoPie activates the device's camera to enable the photo capture feature.



2. Arduino

To facilitate recognition of inserted cards, we employed the MFRC522 RFID module (RFID reader and cards).

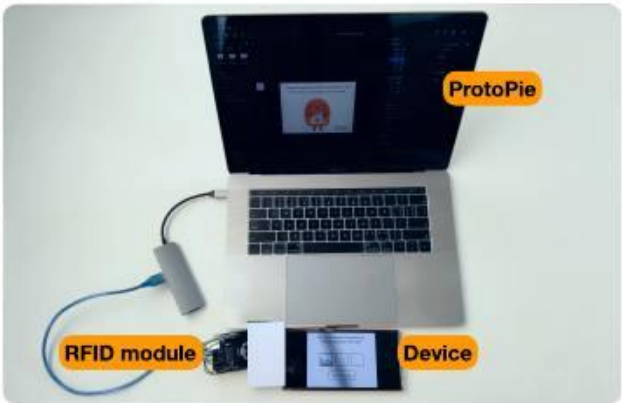
The wiring diagram for the Arduino setup is shown beside, and detailed code can be found in our project's folder.



3. ProtoPie-Arduino Communication

The communication environment utilizes ProtoPie's ProtoPie Connect functionality. This allows data read by the RFID module to be transmitted via serial port to ProtoPie, triggering specific interactions in the prototype.

For instance, in our example, the card information read by the RFID module is sent to ProtoPie as a message labeled "DINOTYPE." If the DINOTYPE equals "Trex," the interface displays a confirmation that the Trex card was successfully inserted.



Message	Message	Value (optional)	Send	
Time	Message	Value	Pin	Source
15:41:08:436	DINOTYPE	Hedro	-	/dev/tty.usbmode...
15:41:03:832	DINOTYPE	Trice	-	/dev/tty.usbmode...

4. Digital Kiosk

To facilitate recognition of inserted cards, we employed the MFRC522 RFID module (RFID reader and cards).

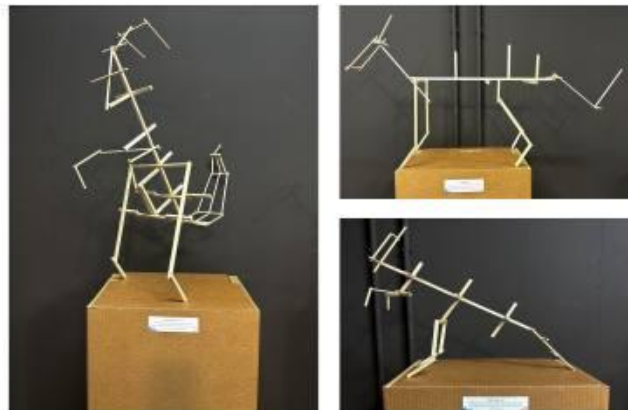
The wiring diagram for the Arduino setup is shown beside, and detailed code can be found in Appendix, page X.



5. Physical Diorama

To facilitate recognition of inserted cards, we employed the MFRC522 RFID module (RFID reader and cards).

The wiring diagram for the Arduino setup is shown beside, and detailed code can be found in Appendix, page X.



Digital Prototype

1. Start

This starting page is designed to create a welcoming experience with our friendly mascot--ECCO.

The cheerful character and "Hi" greeting establish an emotional connection, while the clear "Tap to Start" prompt ensures ease of use, making it intuitive and inviting for users.



2. Inserting cards

Use an interactive "card insertion" visual prompt and animation that turns each card into a story character. This simple yet playful approach keeps users actively involved from the beginning.

The on-screen feedback ("I inserted it") ensures users feel in control.



3. Recognize the card

Confirm successful recognition with clear feedback and give users a choice: keep adding more or start the story. This transparent flow reinforces a sense of control while smoothly guiding the next action.



4. Main story line

Present the central storyline with minimal text, vivid illustrations and lively sound to immerse users in an evolving adventure. We set "Back" button for easy navigation, giving users the freedom to return at any point.



5. Help

Offer an on-screen Help panel so users can pause, restart, or exit the story at any time. This gives them full control over their experience, preventing frustration and enhancing overall usability.



6. Choose adventure

Present branching story options, allowing users to shape the plot. This design choice keeps them actively engaged, granting a sense of ownership and enhancing the interactive narrative experience.



7. Turn on tools

Integrate a tool-activation step (e.g., turning on a compass or camera) to encourage active exploration. By letting users trigger and utilize tools, they remain engaged in uncovering clues and advancing the story.



8. Compass

Integrate a real, functional compass feature that guides users to the story's next clue. Aligning the pins with the "N" marker encourages hands-on interaction, blending narrative with real-world exploration for a more immersive adventure.



9. Camera

Integrate the native camera to capture real photos, merging digital storytelling with physical exploration. This makes the experience both immersive and memorable, as users can document their adventure in real time.



10. Guiding conversation

We use ECCO as a lively, animated guide to bridge kids and parents in a playful, charming way. Its childlike tone and adaptable animations create a fun, welcoming atmosphere, keeping users engaged and reinforcing the product's family-friendly identity.



Case Studies

EXHIBITION

- Experienter rather than witness: to be a part of this journey

"Audiences will encounter species that are adapted for the cold—some long extinct, others still alive today—and learn how a changing climate could alter the face of our planet in the future."



EXHIBITION

- A microscopic view of the macroscopic world
- Impersonal perspective

"Follow Bunji, an Australian green sea turtle, on her first solo journey across the open ocean as she meets incredible animals along the way."



EXHIBITION

- "Non-everyday": uncover hidden
- Immersive environment

"Explore nature's hidden realms--including some that are too fast, too small, or too slow for the human eye."



EXHIBITION

- Cross-platform: before and after museum

"Visitors can share the snowflake designed by themselves on social media as a digital keepsake of the exhibition and access it online."



EXHIBITION

- Inclusive: for adults, be a kid again

"Now it's time for the grown-ups to play and be curious. We have events on tap for you to enjoy without any kids in sight."

Beer, wine and drinks available for purchase. All guests must be 21 or older."



EXHIBITION

- Hospital: museum prescription

"Psychiatrists in one of the city's largest hospitals have been able to offer patients "museum prescriptions", a free visit with a few friends or family members to discover one or more of Brussels' cultural institutions."



Museum Case Study

Name	Topic	What is the experience/ What's it about?	How to interact?	Ages	Touchpoints	Special	Your findings (something you think interesting/useful)	Possibilities for different age groups (4-6 & 7-10)	Possibilities for whole museum or focus on one part
Australian Museum	Live shows		Live show of Shaping prehistoric life features Ella , a fun, fast, friendly dinosaur show, bringing the facts about the Tyrannosaurus family through engaging storytelling and song.		4-6 years			Instead of parents, a real paleontologist explains the scientific facts using children's surprising techniques and songs and dolls.	
	Dinosaur		Create your own miniature and stop-motion production prehistoric world with clay, ice and toy dinosaurs. In this hands-on experience, materials will be provided for you to sculpt and build a diorama landscape using clay, rocks, sand and paint. Dive deeper into the world of filmmaking by mastering the basics of stop-motion production using iPads to create fluid motions and capture dynamic scenes.		5-8 years (kids, no filmmaking); 9-12 years (teenage); 13-16 years (teenage);				
	Climate solutions	Workshop/Interactive installation	Scientist for a Day: Climate Solutions Come behind the scenes to meet some of the Australian Museum's scientists working on climate solutions and the important work they do to help protect our planet. Scientists will guide you through the projects they are working on. The Scientist for a Day program is a full-day social and educational workshop for young minds, designed to show kids what it's like to work in different areas of science.		9-12 years (teenage);				
	Geography	Workshop	Photography Workshop For Kids & Teens Join us for a tour of the Museum's current special exhibition, 2023 Australian Geographic Photographer of the Year, a South Australian Museum exhibition. For kids, introductory photography workshop. Enjoy fun photography challenges set to spark your creativity. For teens, elevate your photography skills. Designed for teens with either beginner or advanced photography knowledge, bring nature into focus as you enhance your skills and creativity.		5-8 years (kids); 9-12 years (teenage); 13-16 years (teenage, some photography techniques);				
American Museum of Natural History	Nature view and life	Immersive experience	Invisible Worlds Explore nature's hidden realms , and discover how all life on Earth is connected. Invisible Worlds explores networks of life at all scales—including some that are too fast, too small, or too slow for the human eye. At key moments, you'll become part of the story as your movements affect the projections around you.		All ages			1. Focusing on the invisible life that people can not experience in the daily life with their eyes. 2. Connect different life together as a story and show the relations and how they are connected, using people body as a guidance to exchange water between living beings.	
	Marine	2D documentary	Turtle Odyssey Follow Bang, an Australian green sea turtle, on her first solo journey across the open ocean as she meets incredible animals along the way.		All ages		Storytelling (perspective)		
	Gems and Minerals	Conventional exhibition	Ice Color: An Exhibition of Hip-Hop Jewelry See stunning jewelry pieces that trace the history of hip-hop from the 1980s to today. It celebrates hip-hop's cultural influence through stunning jewelry worn by some of its iconic stars.		All ages		Storytelling (perspective)		
	Ice Ages, Climate change	Comprehensive interactive exhibition (educational intention)	Planet Ice: Mysteries of the Ice Ages Through engaging exhibits, multimedia and some "magical" moments, audiences will encounter species that are adapted for the cold—some long extinct, others still alive today—and learn how a changing climate could alter the face of our planet in the future.		All ages				
The Wolf Museum of Exploration	Exploration, education, Program: STEAM activities (STEM+ART)		Toddler Tuesdays Exploring monthly themes, special activities, and modifications to our exhibits to make them more accessible to little ones. Take this time to play, discover and interact with exhibits alongside your toddlers. Cooperated with UCR's Child Studies Lab, facilitate an activity they have designed for toddlers based on scientific research.		For families				
	Adult activities	Program (Happy Hour, Afterparty, Seminar...)	Be a kid again Why should kids have all the fun? Now it's time for the grown-ups to play and be curious. We have events on tap for you to enjoy without any kids in sight .		21+				
Smithsonian National Air and Space Museum (Outstanding Immersion experience)	Spaceflight	Comprehensive interactive exhibition	At Home in Space An immersive, highly interactive exhibition. At Home in Space takes visitors along a 40+ year journey of learning how humans can live and work in space comfortably and venture beyond Earth orbit.		All ages				

link: https://docs.google.com/spreadsheets/d/19otQNfGK4dY_wqCst7mabaF_p3XzupbchefeJFhhyg/edit?gid=0#gid=0

Interview Questionnaire in the museum

1/14/25, 5:05 PM

Museo di Storia Naturale di Milano

Museo di Storia Naturale di Milano



Basic information

1. How old is the parent?/How old are the parents?

2. How old is the child/How old are the children?

<https://docs.google.com/forms/d/1K9oQx8e3f5hp7A3k3e3ISn-JejV6KpaMIMgGhwdPfA4/edit>

1/16

link:<https://docs.google.com/forms/d/1K9oQx8e3f5hp7A3k3e3ISn-JejV6KpaMIMgGhwdPfA4/edit#responses>

Feedback Board- interview in museum

Quale parte del museo non ti piace? Perché?
Which part of the museum you don't like? Why?

For parents

1. Mineralogia Mineralogy 	2. Paleontologia Paleontology 	3. Storia Naturale Dell'uomo Natural History of Human 
4. Zoologia Degli Invertebrati Invertebrate zoology 	5. Entomologia Entomology 	6. Zoologia Dei Vertebrati Vertebrate zoology 

Qual è la parte più interessante del museo? Perché?
What's the most interesting part in the museum? Why?

For parents

1. Mineralogia Mineralogy 	2. Paleontologia Paleontology 	3. Storia Naturale Dell'uomo Natural History of Human 
4. Zoologia Degli Invertebrati Invertebrate zoology 	5. Entomologia Entomology 	6. Zoologia Dei Vertebrati Vertebrate zoology 

Feedback Board- interview in museum

Ti piace il museo?
Do you like the museum?



Si, mi piace
yes, i like it



non mi piace
no, i don't like it

Ti piace il museo?
Do you like the museum?

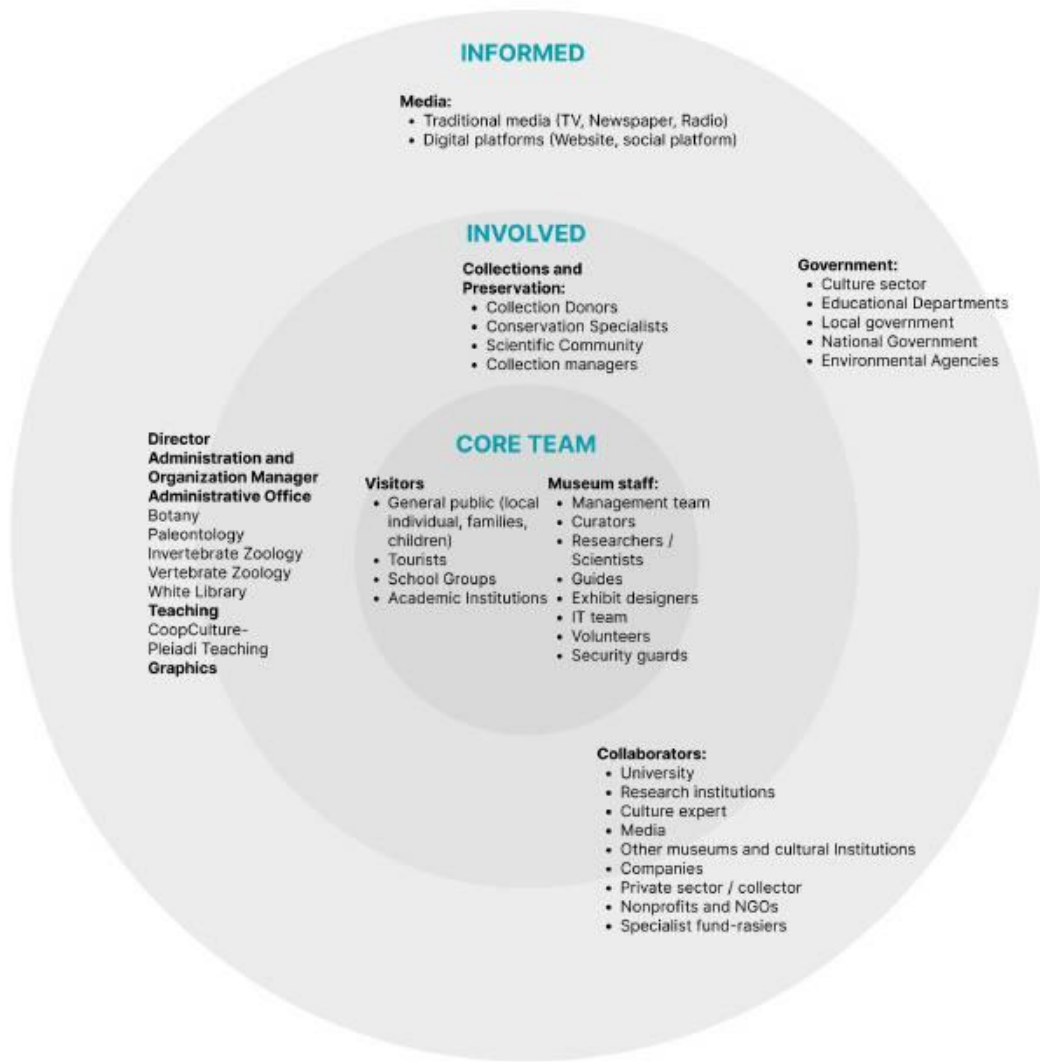


Si, mi piace
yes, i like it

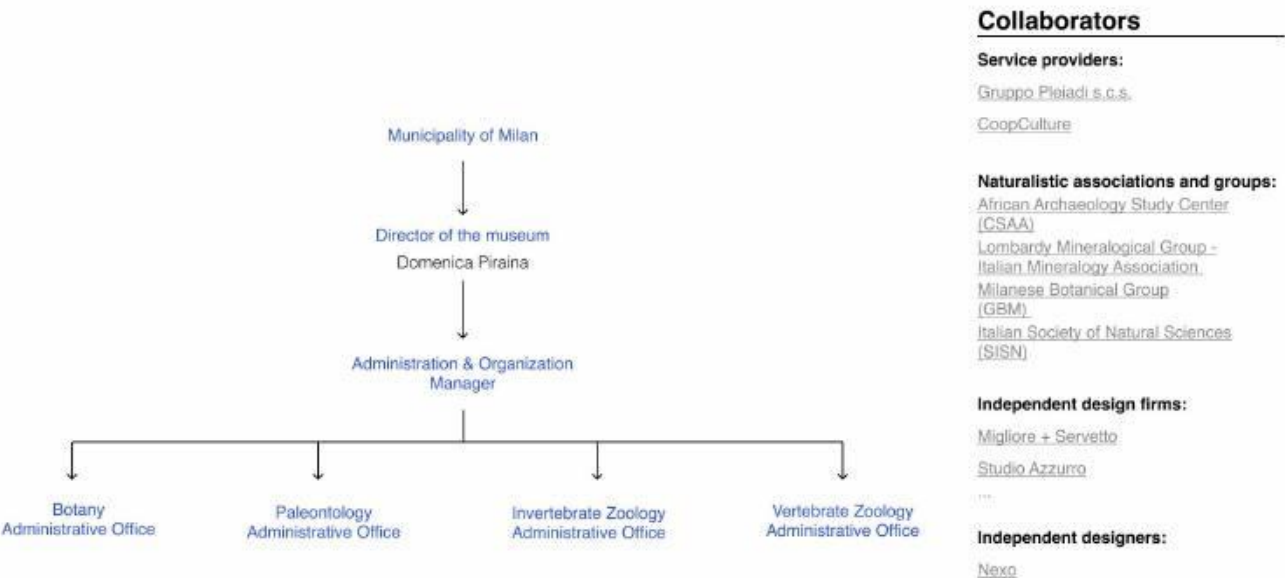


non mi piace
no, i don't like it

Ecosystem map



Organization chart



User Journey Map For Interactive Table (Idea 1,2) , User Experience Flowchart
please click the like to see the detailed map and chart.

https://drive.google.com/drive/folders/1IHCqn2vze1rME9FWxdtlr9cR9lr8jNxW?usp=drive_link